



POOJAN SHAH

RESEARCHER

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RESEARCH INTERESTS

Research-focused computer scientist working at the intersection of large-scale machine learning, algorithms and optimization, and quantum computing.

EDUCATION

Indian Institute of Technology Delhi (IIT Delhi) New Delhi, India
Bachelor of Technology (B.Tech.) in Computer Science and Engineering 2022 – 2026 (expected)

- Cumulative GPA: 9.86 / 10.00
- Research Advisors: Prof. Ragesh Jaiswal, Prof. Rajendra Kumar
- Research projects: intrinsic dimension estimation; improved classical and quantum algorithms for large-scale data clustering

University of Waterloo Waterloo, Ontario, Canada
Visiting Undergraduate Student, Faculty of Mathematics Winter 2025
• Grade: 94.20 / 100.00

PUBLICATIONS

1. **P. C. Shah**, R. Jaiswal. Quantum (Inspired) D^2 -Sampling with Applications. International Conference on Learning Representations (ICLR), 2025. [OpenReview].
2. **P. C. Shah**, S. Agarwal, R. Jaiswal. Fast k -means seeding under the manifold hypothesis. International Conference on Machine Learning (ICML), 2026 (To Appear). [arXiv]

RESEARCH EXPERIENCE

Wadhvani Institute for Artificial Intelligence | Mumbai, India Nov 2025 – Present
• Supervisor: Prof. Makarand Tapaswi
• Research on self-supervised learning from large scale unlabelled data for newborn anthropometry

Atlas Research | Dubai, United Arab Emirates May 2025 – Jul 2025
• Developed optimizations for large-scale financial exchange Application Programming Interface (API) systems.
• Implemented Bayesian optimization pipelines for hyperparameter tuning.
• Conducted large-scale feature selection and predictive modeling for crypto data.

Centre for Quantum Technologies, NUS | Singapore May 2024 – Jul 2024
• Supervisor: Prof. Rahul Jain
• Studied cryptographic primitives based on pseudorandom quantum states and their complexity-theoretic guarantees.

SCHOLARSHIPS & FELLOWSHIPS	• Amit and Deepali Sinha Foundation Fellowship Full undergraduate funding awarded to the top 5 students at IIT Delhi 2022 • IIT Delhi Merit Award Awarded to Top 7% of CSE department (all semesters) 2022–2025 • Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship National science fellowship bestowed by the Indian Institute of Science (IISc), Bangalore; All India Rank 112 2022 • National Talent Search Examination (NTSE) Scholarship National-level merit scholarship administered by NCERT 2020 • ICLR Travel Award: Singapore 2025 • Cornell–Maryland–Max Planck Research School (CMMRS) MPI-SWS, Germany (selected participant) 2025 • Google DeepMind Research Symposium: Selected participant 2024 • ACM India Annual Event: Invited for lightning talk and research presentation 2025
SELECTED ACADEMIC ACHIEVE- MENTS	• JEE Advanced: All India Rank 67 among 250,000 candidates 2022 • JEE Main: All India Rank 248 among 1,000,000 candidates 2022 • IBM Quantum Excellence Badge: Qiskit Global Summer School 2023 • OPhO: World Rank 3 among 600 teams 2021 • Fyziklani: 1st place, international physics competition, Charles University, Prague 2021 • National Standard Examination in Physics (NSEP): Top 1% nationwide 2021 • Indian National Chemistry Olympiad (INChO): Top 49 nationally 2021
TALKS & PRE- SENTATIONS	Quantum and Quantum-Inspired Algorithms for Data Clustering. <i>CS Group Meetings, CQT, NUS.</i>
	Quantum Machine Learning Without Any Quantum. <i>Theory Seminar, CSE Department, IIT Delhi.</i>
TEACHING	
	Head Teaching Assistant for Quantum Computing , Winter 2026, IIT Delhi. <i>Created problem sets, led tutorial sessions, and mentored undergraduate students</i>
SKILLS	
	Programming: Python, C++, R
	Machine Learning: PyTorch, scikit-learn
	Scientific Computing: NumPy, pandas
	Research Tools: LaTeX, Git, Shell
	Coursework:
	<i>Computer Science:</i> Data structures, Algorithm Design, Digital System Design, Computer Architecture, Computer Networks, Operating Systems, Parallel & Distributed Systems, Machine Learning, Principles of Artificial Intelligence, Quantum Computing, Cryptography <i>Mathematics:</i> Real & Complex Analysis, Harmonic Analysis, Linear Algebra, Abstract Algebra, Differential Equations, Measure Theory & Probability, Stochastic Processes, Convex Optimization <i>Physics:</i> Classical Mechanics, Electrodynamics, Quantum Mechanics, Quantum Information Theory, Signals and Systems